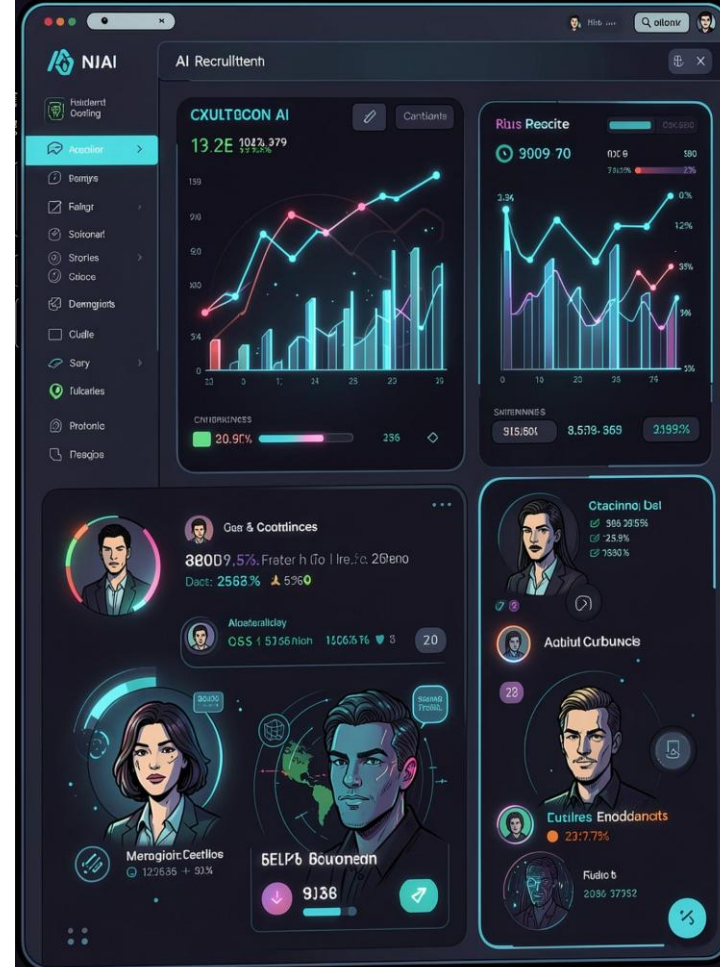


RECRUITAI

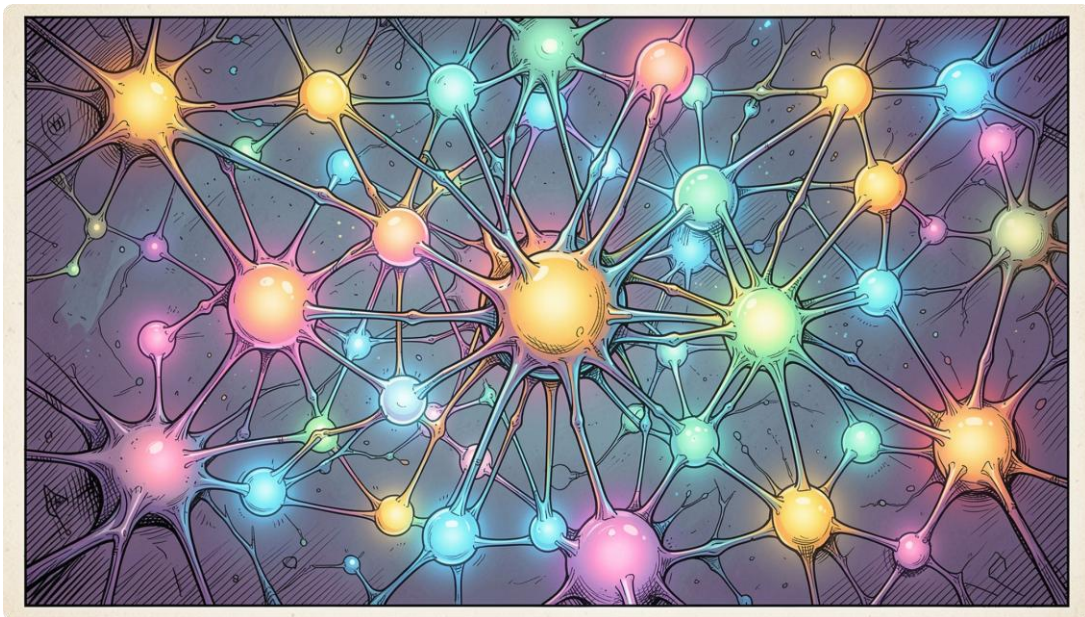
AI-Powered Candidate Ranking System

Traditional recruitment relies on keyword matching — overlooking qualified candidates who use different terminology. RecruitAI understands **intent**, evaluates candidates **holistically**, and delivers a **trustworthy ranked shortlist**.

Team Leader : Sachin Singh



Solution Overview



What is RecruitAI?

A hybrid AI candidate ranking system combining **semantic understanding** with **feature engineering** to evaluate candidate-job fit.

- Converts JD and profiles into dense vector embeddings using Sentence Transformers
- Evaluates structured recruiter signals: skills, experience, behavior, career progression
- Produces recruiter-quality, explainable rankings at scale

What Makes RecruitAI Different



Semantic Understanding

Understands meaning, not just keywords — identifies relevant candidates even when exact terms are absent.



Structured Recruiter Signals

Evaluates skills, experience, GitHub activity, recruiter engagement, and behavioral indicators.



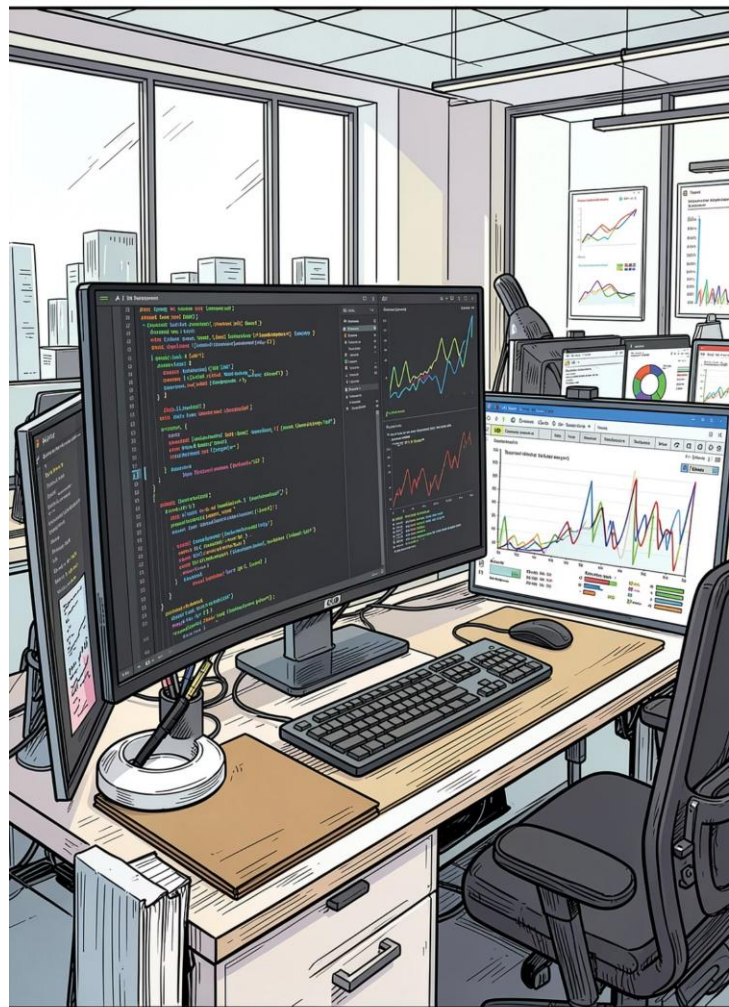
Explainable Scores

Generates interpretable ranking scores for every candidate, enabling transparent recruiter decisions.



Scalable by Design

Efficiently processes large candidate datasets with pre-computed embeddings and single-pass scoring.



JD Understanding & Candidate Evaluation

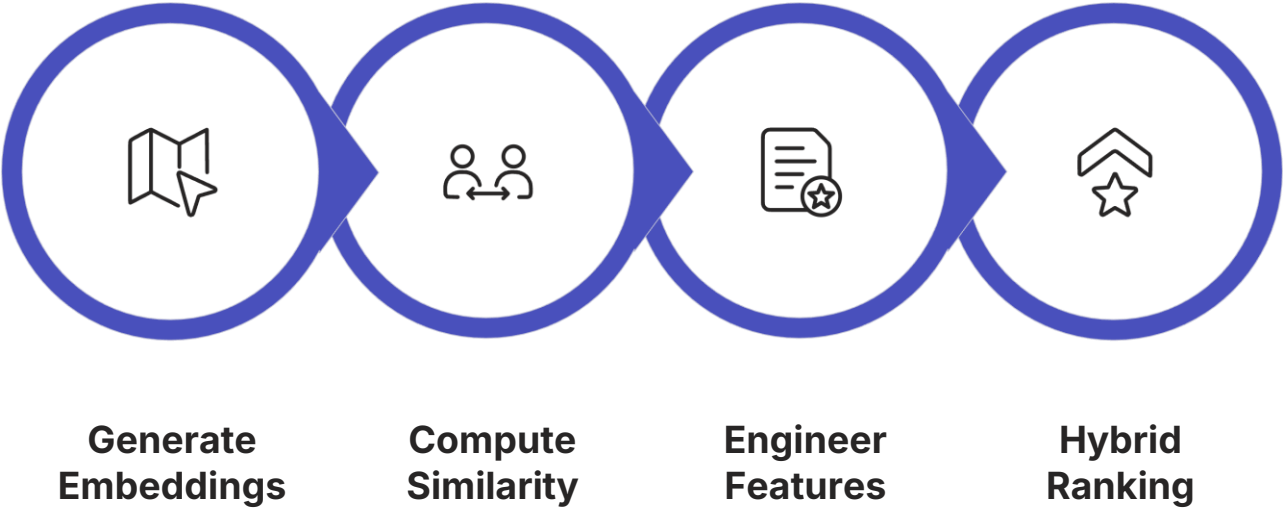
Key JD Requirements

- AI, ML, NLP & LLMs
- Search, Ranking & Retrieval Systems
- Recommendation Systems
- RAG & Vector Search
- Production ML Systems
- Strong Software Engineering

Candidate Signals Evaluated

- Technical skills & years of experience
- Career history & current role relevance
- Search/AI domain expertise
- Recruiter engagement & interview rate
- GitHub activity & open-to-work status
- Semantic similarity with the JD

Ranking Methodology



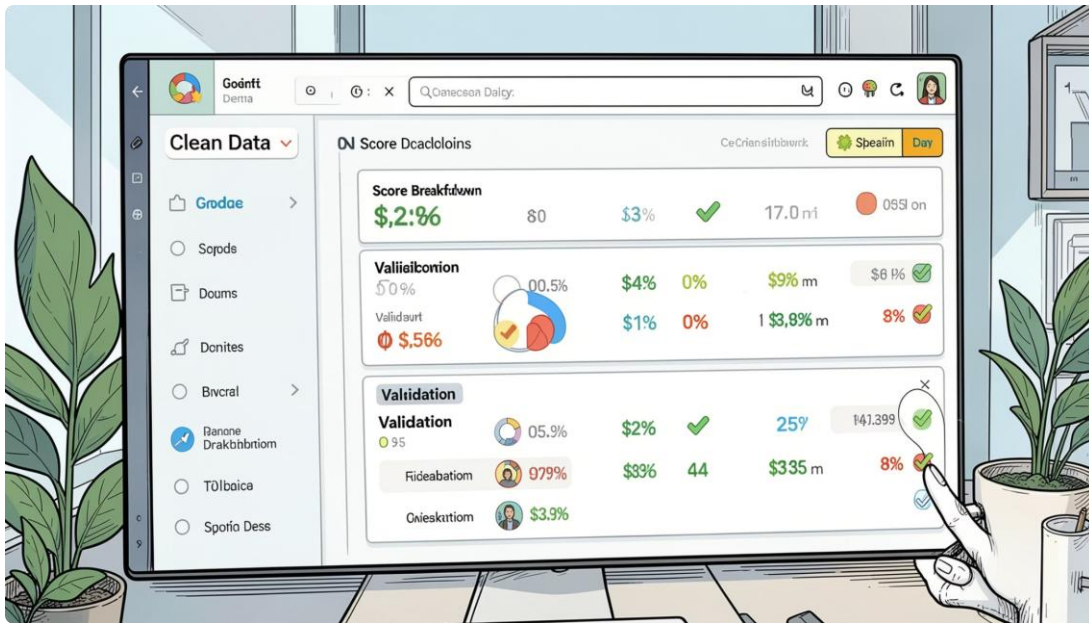
RecruitAI's ranking pipeline flows through four stages — from raw text to a final ranked shortlist — combining semantic embeddings with structured recruiter signals.

Final Ranking Formula

Signal	Weight
Semantic Similarity	25%
Search & AI Expertise	20%
Skill Score	15%
Experience Score	10%
Founder/Ownership	10%
Behavioral Score	10%
Role Relevance	10%

Penalties applied for irrelevant roles. All scores normalized 0-100.

Explainability & Data Validation



Transparent Score Breakdown

Every candidate receives an interpretable breakdown across 7 dimensions — enabling recruiters to understand **why** a candidate received their ranking.

Preventing Hallucinations

All scores are computed directly from **structured candidate data** and semantic similarity. No external information is introduced during ranking.

Handling Inconsistent Profiles

- Missing values and null fields handled gracefully
- Irrelevant job titles receive automatic penalties
- Structured data validated before scoring

End-to-End Workflow

01

Input Job Description

03

Generate Job Embedding

05

Build Candidate Profiles

07

Calculate Feature Engineering Scores

09

Rank All Candidates

02

Extract Job Requirements

04

Read Candidate Dataset

06

Generate Candidate Embeddings

08

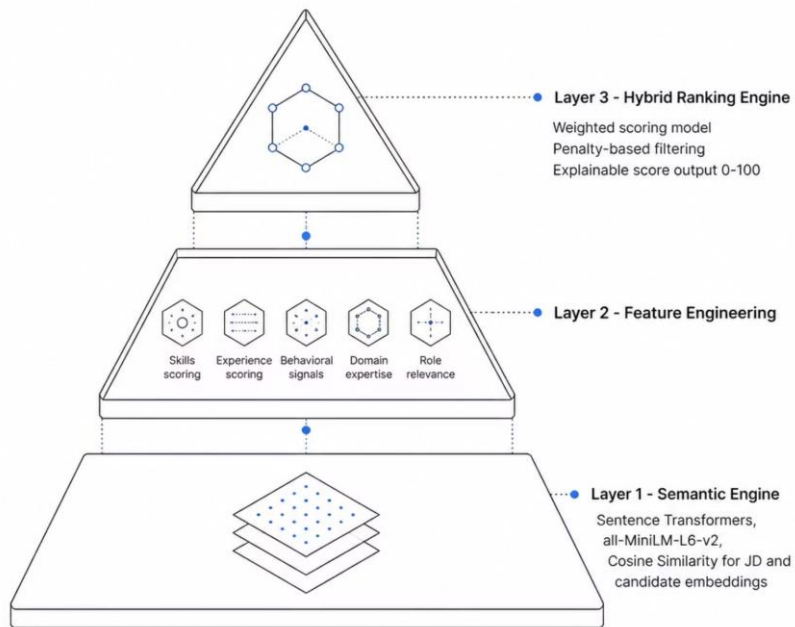
Compute Hybrid Ranking Score

10

Generate Submission CSV

The pipeline processes the Job Description once, then scores all candidates in a single efficient pass — from raw input to competition-ready output.

System Architecture



How It All Connects

RecruitAI combines **semantic search** and **feature engineering** in a unified architecture.

- Sentence Transformers capture contextual similarity between JD and candidate profiles
- Structured recruiter signals improve ranking precision beyond pure semantics
- The Hybrid Ranking Engine integrates all signals into a single explainable score

Every score is traceable, reproducible, and recruiter-interpretable.

Results & Performance

100K

Profiles Ranked

Successfully processed and ranked
100,000 candidate profiles.

25%

Semantic Weight

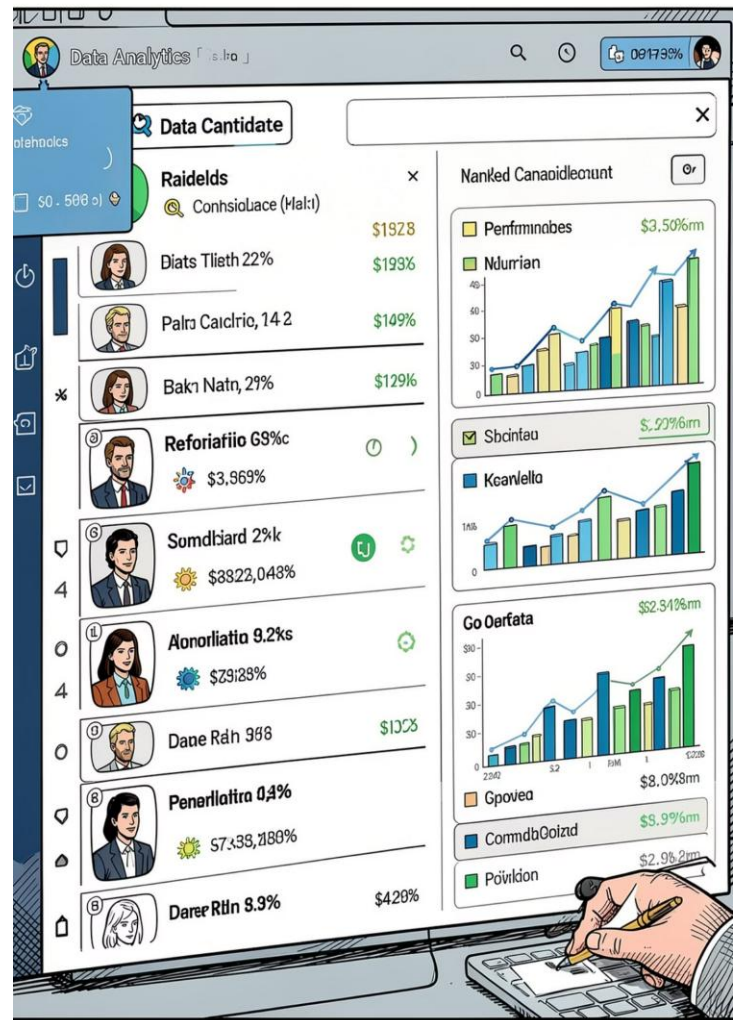
Largest single signal weight — semantic
similarity drives ranking quality.

1

Pass Processing

Single-pass candidate processing with
pre-computed JD embedding.

- Top-ranked candidates consistently belong to AI, Search, Retrieval, and Recommendation domains
- Semantic similarity significantly improves relevance beyond keyword matching
- Lightweight MiniLM model ensures efficient, scalable computation
- Competition-compliant submission.csv generated successfully



Technologies & Submission Assets

Technology Stack

Python

Core programming language

Sentence Transformers

all-MiniLM-L6-v2 for embeddings

Scikit-learn & NumPy

Feature engineering & computation

Git & GitHub

Version control & collaboration

Submission Assets

- Complete source code with README
- `requirements.txt` for reproducibility
- Architecture diagram & presentation PDF
- Ranked `submission.csv`

Project Features

HYBRID AI RANKING

SEMANTIC MATCHING

EXPLAINABLE AI

SCALABLE

COMPETITION READY